

papers. To discover the older literature, I would then examine citations in these papers, inferring the important sources from the frequency and nature of the references to them. The big advantage of this method is that it gives you the citation structure of the field and some feel for how things have developed. A related technique searches in the opposite direction, and search engines are very helpful here. Identify a key paper on the topic published, say, 20 years ago and then look at the papers published since then that cite this paper. Sometimes, you pick up papers you would otherwise miss because they were published in unusual places or lacked obviously linked keywords or titles.

While on the subject of computers, it is very important in the information age to understand the difference between knowledge and information. What is accessible by computer and, indeed, what is published in the journals is information. Knowledge is something that has to be constructed in the mind of the expert reader. This is what scholarship is about. Information is, these days, instantly accessible, but knowledge still takes years of dedicated study to acquire. Imagine that a freak accident wiped out an entire field of experts on a subject while all were attending a conference. How long would it take to reconstruct expertise in the field so that research could once again progress? It would probably take many years, despite the fact that their research was all published. To take another example, what do producers of science documentaries for television programmes do when they are researching their subjects? They talk to the experts rather than trying to read the journals. Quite rightly, as that is the only place that knowledge is to be found – inside the heads of the scholars.¹

The purpose of academic research is to advance the collective knowledge of the discipline. Although the process also advances the knowledge of the individual, this is not sufficient. For example, if someone in ignorance of the literature ‘discovers’ something that was already known to the field, then they have not advanced knowledge in the sense that I am talking about here. However, things are really more complex than this. Research must be *contextualised* in its presentation. The context is the knowledge of the relevant topic prior to conducting your piece of original empirical research. In a journal article, this context is provided by the introduction, which includes a highly selective and focussed literature review. The discussion of the journal article relates the new findings to the context created. Hence, while scholarship is not sufficient to define research, it is necessary to conduct and report it. I will return to the writing of journal articles in Chapter 7. For now, I will consider the writing of a literature review as an object in itself. This might be a review written for the introductory chapters of a PhD thesis, for a review article, or for a book chapter or an entire book.

Writing a literature review

One of the reasons that experts frequently disagree is that no two people can ever have the same knowledge base. I am fortunate enough to have worked extensively with colleagues who shared a lot of my interests in thinking and reasoning. One of the reasons that research groups may be more productive than individuals (see

Chapter 6) is that there is a good deal of shared knowledge. Many of the papers that I have read and thought about will be known to my colleagues, so we do not have to waste a lot of time and effort establishing background knowledge when we want to discuss issues of interest to our research. This is, of course, facilitated by regular workshops and seminars. However, it is evident that no two people can have read exactly the same literature in the same order, nor have brought to it the same prior knowledge and attitude. If I am right in my assertion that knowledge, as opposed to information, exists only inside the heads of the human experts, then this implies that the knowledge acquired by science is not entirely objective and universal. It is to an extent subjective and individual.

Anyone involved in cutting edge research must be aware that the state of knowledge in their field is fuzzy and ill-defined. Leading authors promote conflicting theories and impose differing interpretations on the research findings reported in the literature. There *is* no definitive state of knowledge. It is important to be aware of this when writing a literature review. All such reviews, however scholarly, are constructions of the authors who write them. All reviews are by necessity selective in the citation and discussion of the massive amount of information that is available in the journals. No two authors would make the same selection or structure the material similarly, nor interpret it in precisely the same way. However, this does not mean that, as a scholar, you have licence to write anything you want. Some reviews will satisfy their academic critics (PhD examiners, journal reviewers, etc) and some most certainly will not.

The subjective aspect of a literature review became very clear to me when writing a review of my own field for the premier review journal *Psychological Bulletin* (Evans, 2002). The review encompassed a potentially huge amount of literature, since it covered the psychology of deductive reasoning over a 40 year period of work. The review was historical by its nature and purpose. The point was to argue that the central paradigm by which deduction was studied was based on a framework (logicism) that made sense when the paradigm was developed in the 1960's but that subsequent developments had made this methodology much more questionable as a method for studying human reasoning. The idea behind this paper was originally presented as a keynote address at a major conference and the reaction of colleagues to this was sufficiently positive to encourage me in the view that the field might be ready to hear this argument.

Note the references here to *purpose* and *argument*. Science is not the mere collection of facts, and nor is a review the mere summary and recording of evidence. You must always write to a purpose and indeed *read* to a purpose, as I shall explain below. However, the problem in writing a review for a journal, especially one as influential as *Psychological Bulletin*, is that you have to satisfy referees who are invariably other experts in the same field. Their view of the field will inevitably be different from yours but they know that your article, once published, will have a lot of influence on how other people – especially students and young researchers – will think about the topic. The first thing you have to do is impress these reviewers with your scholarship. You must appear to be (and actually be!) both widely read and possessed of a depth of understanding of the subject, even if it differs

from their own. The second requirement they will have is that you are *fair* to other authors in the field – especially them! I will have a lot more to say about referees and publication in Chapter 7.

Literature reviews can be vastly influential. A good example is the book *Cognitive Psychology* (the first of many with that name), written by Ulrich Neisser (1967). Prior to publication of this book, the term ‘cognitive psychology’ was not in general use. Although the anti-behaviourist cognitive revolution had been in full swing for the previous 10 years or so, its replacement did not yet have a clear identity or name. Until Neisser’s book was published, that is. From an objectivist viewpoint, Neisser did not invent the field – the research was already there in the literature, together with theoretical accounts reflecting the information processing approach. From the subjectivist point of view I am arguing, however, Neisser did in a sense contribute directly to the invention of the field. In this brilliant review, he constructed and identified the study of cognitive psychology. It influenced and inspired an entire generation of researchers, including me. The book was published while I was a final year undergraduate at UCL. It remains, to this day, the only psychology book I have read from cover to cover *twice*.

Structuring and organizing the review

Let us assume that you have done your literature search and have a pile of reprints sitting on your desk, or more likely these days, a folder full of pdf files on your computer. Your task is to read these papers and then write a review of them. How should you go about this? Most PhD students, in my experience, try to read the papers end to end, in no particular order. This approach is not only highly inefficient, but typically leaves the student exhausted and confused. Let us assume that most of these papers are journal articles reporting new empirical findings. (You certainly do not want to read someone else’s review of the same field just before writing yours.) Each paper will tell its own story from the viewpoint of its own authors. Even though the papers fall into the same broad area, each is likely to have an introduction that reflects a somewhat different set of literature and a discussion that interprets findings within the authors’ favoured theoretical perspective. Even papers that are presenting similar findings may frame and interpret them in quite different ways. Hence, if you read each paper within the structure that the authors provide, the total picture will be confusing and hard to integrate.

The key concept in literature review is *structure*. The review itself will need a structure, which can be done in several ways, as I discuss below. The sooner you can develop an outline of this structure, the better, as this will greatly assist you in the task of reading the literature. Hence, you need to do a preliminary reading that is relatively quick and superficial in order to decide how you wish to structure the review, and then do your detailed reading at a point where you have the main headings and subheadings. At this point, let me say a few words about depth of structure. Some PhD theses are presented using the numbering system of heading, which their authors take as a licence to embed to any level of depth. It is not unusual to encounter a section headed 2.1.3.4, or whatever. This is bad

practice – do not do it! The point about your structuring of a review is that it must be reader-friendly. At all times, the reader must have your overall structure in mind and be able to follow where they are within it.

I strongly recommend an approach that allows only three levels: chapter, main heading and subheading. The first comes into play only when writing a thesis or a book. So a single paper, such as a review article, should permit only main and subheadings – just two levels. I have never encountered material that could not be structured in this way (although I should note that a distinguished commentator on my draft disagreed!). Keep in mind that the structure is something that you impose on the literature. You do it in order to make your assessment of it as easy as possible for the reader to follow and understand. Sections also have to be of an appropriate length that falls within the span of apprehension of the reader. As a general rule, I would recommend sections (or subsections) no longer than 2,000 words when reviewing technical material. If it needs to be longer, then divide it further.

The structure that you choose will, however, depend to some extent on the material that is being reviewed. I shall illustrate this with reference to a couple of my own publications. In the high level textbook on deductive reasoning that I wrote in collaboration with two colleagues (Evans, Newstead, & Byrne, 1993), we found that some chapters were best structured by the task that were used to study the phenomena: this worked well for Chapter 2 (conditionals) and Chapter 4 (disjunctives). But the same approach could not be used for the chapter (7) on syllogistic reasoning, as in this case, there is essentially only one task employed but a dozen different theories proposed to explain the findings. So that chapter was structured by theories, with major headings for related classes of theories and subheadings for specific accounts of the phenomena.

The second example is my review of dual process theories for the *Annual Review of Psychology* (Evans, 2008), my most successful paper in terms of citations. My remit was to cover such theories in three different fields: reasoning, decision making and social cognition. The second part of the paper does this, with a heading for each. However, my ambitions for the paper were more than to simply describe the major papers. I wanted to show a number of strands of common thinking that underlie dual process theories in these different fields and to identify common problems with the underlying thinking. So I started by listing similar attributes in the numerous theories of this type,² which I organized into four clusters: consciousness, evolution, functional characteristics and individual differences. Hence, the first part of the review had four subsections, allowing me to discuss the common issues with each of these clusters. Note that I did this *before* describing the theories in each field. Of course, my own understanding reflected in Part One emerged from reading the literature described in Part Two. But the reason for ordering it this way was to provide the reader with a frame within which more general issues were understood first and specific theories seen in that context.

It is important to identify as early as possible the structure that will suit your particular material. There is no uniquely correct way to do this, but a good way

will reflect the various constraints that I have indicated: (a) it will reflect the nature of the material being reviewed, (b) it will enable you to make sense of the material and convey this understanding to the reader and (c) it will permit a two-level heading and subheading organization. Much of your detailed reading will be done *after* this structure has been decided. So how do you go about choosing the structure? This depends on whether you have some prior knowledge of the topic or whether you are coming to it cold. In either case, I would avoid reading other people's reviews of the topic, as you do not want to inherit their method of structuring the literature. If you are new to the field, then try to identify a subset of your mountain of reprints or pdf files that are the *key papers* in the field. Key papers are the ones that most influence research on the topic. Clues to which papers are key ones include the rate at which they are cited and the prestige value of the journals in which they are published. (Papers published in top ranked journals tend to be cited a lot, even if they are not particularly good.)

It is worth reading key papers in full, but with the others at this stage, you can just read the abstracts or skim the introductions in order to classify them. You should end up with heaps and sub-heaps of reprints (or folders and sub-folders of pdf files) corresponding to the headings and subheadings that you will use in your review. In this way, you can save much of your detailed reading to be done immediately before you write your 2,000 word section. This is very efficient, since if you read everything to start with, you will forget a lot and end up re-reading at the point of writing in any case. Even at the detailed reading and writing stage, it may not be necessary to read all the papers in full. It all depends upon the purpose to which you are reading and writing.

The writing of empirical journal articles is an art about which I will have much to say in Chapter 7. However, the basic schema has a lot in common with certain types of literature review. You have an introduction that lays out the basic context for the research; an experimental section that reports collections of new data and, finally, a discussion that makes sense of the new data in the context provided by the introduction. When I wrote my first major review – a textbook on deductive reasoning (Evans, 1982) – I already had about 10 years' experience of writing journal articles. What excited me about the review process was the much larger database of studies to interpret and discuss. I thought of each chapter as a big journal article, but instead of an experimental section with just two or three experiments of my own to consider, I had perhaps 20 to 40 studies on a particular topic to discuss. So what I did was read primarily the method and results sections of these journal articles, with minimal attention to the introduction and discussion sections. Where appropriate, related studies can be tabulated together, as we did quite frequently in the 1993 re-write of this textbook (Evans et al., 1993). You might even want to adopt the modern practice of meta-analysis (Rosenthal, 1993), where statistical techniques are used to provide inferential statistical analysis of a set of different studies.

Whichever technique you use, the point is that you are assembling data and then interpreting it. Reading the outer sections of the papers will hinder this process more than help it, as all the authors will have different ideas. It is *your* interpretation of the data that the review should reflect. However, that is not to say that

introduction and discussion sections are a waste of journal space! You might, for example, be structuring a review section about theories, ideas and issues. In this case, the outer sections of the papers would be much more the focus of your attention than the inner ones. Again, it all depends on the purpose to which you are reading and writing.

The origin of ideas

Where do ideas for new research come from? This is an issue that philosophers of science have traditionally shied away from, declaring it to be primarily a psychological issue. There is much discussion to be found among the philosophers about how hypotheses and theories should be tested, some of which I will consider in Chapter 8. But where do they come from in the first place and to what extent does this depend upon scholarship? Recall that Peter Wason, who placed great emphasis on creativity in science (Wason, 1995), thought that reading and scholarship interfered with this process.

I think that we must acknowledge that creative researchers like Wason who lead thinking on their subjects are the exception rather than the rule. It is true that his work mainly reflected his thought about the issues rather than his scholarly appreciation of the work of others. A general problem is that the few sources of psychological studies of scientists and their thinking have tended to focus on great scientists and geniuses. Thus we have, for example, Wertheimer's (1961) account of the thought process of Albert Einstein as he grappled with the theory of relativity – fascinating stuff, but hardly helpful for the average PhD student. There are some systematic studies of great scientists to be found in the literature, such as the analysis of Alexander Graham Bell's diaries published by Gorman (1995). More useful, perhaps, are naturalistic studies of scientific research groups such as those conducted by Kevin Dunbar and his colleagues (Dunbar, 2002), but these generally provide more insight into reasoning processes than creativity.

'Normal' science is paradigmatic in the Kuhnian sense and hence does rely on scholarship to a large extent. To advance knowledge, we must know how far it has progressed in the first place. At the cutting edge, however, nothing is clear cut and decisive, certainly not in psychology. For every answer we find, we usually discover three new questions. In many ways, doing research is like reading a good novel. You get absorbed in the plot and eager to read the next chapter. Then you discover the next chapter has not yet been written, so you have to design your own research study to see where the story goes next. A student or early career researcher will usually get ideas from reading the literature, getting interested in an issue and starting to wonder about some aspect that is not yet understood. Or they may be inspired initially by an individual – perhaps a distinguished researcher in the department where they work. If you have the Scientist personality type that I discussed in the Introduction, you may develop a passionate curiosity that only research can satisfy. Established researchers may spend much time answering questions that their own earlier research has opened up, but there are individual differences here to which I will return.